

Bayesian Deep Neural Networks and Non-Identifiability

An annotated survey of the “Big Four” statistics journals — *Annals of Statistics*, *JASA*, *JRSS-B*, *Biometrika*

Special attention to treatments of the **quotient space** induced by weight-space symmetries · Coverage: 2020–July 2026, plus a pre-2020 foundational section · Compiled 23 July 2026

Scope, and an honest warning about it

The three-way intersection requested here — Bayesian *deep* neural networks $\times$ non-identifiability $\times$ the quotient space — is close to empty in the Big Four. That literature lives overwhelmingly in machine-learning venues (NeurIPS, ICML, AISTATS, JMLR). The Big Four papers that do engage it arrive through the statistician’s vocabulary for the same object: *equivalence classes of networks*, *structure-selection consistency*, and *posterior invariance*. Quotienting the parameter space by the permutation, sign-flip and rescaling group is exactly what makes “the” posterior mode, or a marginal inclusion probability, a well-defined target at all.

Completeness caveat. This was assembled by targeted literature search rather than by walking the complete table of contents of each journal issue by issue. Treat it as “everything that could be substantiated,” not a guaranteed-exhaustive census. Every citation below has been checked against the publisher record; the appendix lists the queries that would close the remaining gap.

Tier	Meaning
A — Direct	Explicitly constructs or reasons on the quotient / equivalence class of networks induced by weight-space symmetry.
B — Strong	Explicitly names non-identifiability and ties posterior behaviour to it, without a full quotient construction.
C — Adjacent	Attacks a consequence of non-identifiability (multimodality, local traps, invalid uncertainty quantification) without framing it as such.

Part I — Before 2020: foundational and influential

Included because both papers are load-bearing for everything in Part II, and because the first is the earliest Big Four paper to connect a posterior directly to a neural network’s lack of identifiability.

Biometrika

2001

Paige, R. L., and Butler, R. W. (2001). “Bayesian inference in neural networks.” *Biometrika*, **88**(3), 623–641.

doi:10.1093/biomet/88.3.623

Summary. Develops approximate marginal Bayesian computation for neural networks — Bayes factors for the number of sigmoid terms, predictive densities, and Laplace approximations — after conjugate elimination of the linear parameters yields an explicit posterior on the nonlinear ones.

Non-identifiability link. The abstract states outright that the choice of prior together with an alternative sigmoid produces *posterior invariance in the nonlinear parameter*, which the authors discuss in connection with the lack of sigmoid identifiability — the earliest explicit posterior/non-identifiability link in the Big Four.

Quotient space. Posterior invariance under the symmetry group is precisely the statement that the posterior descends to the quotient; the paper reaches the idea without the modern vocabulary.

Relevance tier: B — Strong (shallow network, not deep)

Journal of the American Statistical Association

2018

Liang, F., Li, Q., and Zhou, L. (2018). “Bayesian Neural Networks for Selection of Drug Sensitive Genes.” *JASA*, **113**(523), 955–972.

doi:10.1080/01621459.2017.1409122

Summary. Casts biomarker discovery from high-dimensional omics data as variable selection through a feed-forward Bayesian neural network, using the network’s universal approximation property to handle an unknown nonlinear functional form.

Non-identifiability link. Variable-selection consistency for a neural network is only a coherent claim once the network’s symmetry-induced redundancy is accounted for; this paper establishes the posterior-consistency footing that the 2022 quotient construction then makes explicit.

Relevance tier: B — Strong (direct intellectual ancestor of the Tier A paper below)

Part II — 2020 to present, by journal and year

Journal of the American Statistical Association

2020 — no qualifying article identified

2021 — no qualifying article identified (the Tier A paper below appeared online in April 2021 but is bound in the 2022 volume)

2022

Sun, Y., Song, Q., and Liang, F. (2022). “Consistent Sparse Deep Learning: Theory and Computation.” *JASA*, **117**(540), 1981–1995.

doi:10.1080/01621459.2021.1895175

Summary. Proposes a frequentist-like procedure for learning sparse deep neural networks and justifies it in a Bayesian framework, establishing posterior consistency under a mixture Gaussian prior, variable-selection consistency, and asymptotically optimal generalization bounds for networks with at most $O(n/\log n)$ connections.

Non-identifiability link. Over-parameterized DNNs are massively non-identifiable, so the authors do not attempt inference on raw weights at all; the paper’s structure-selection theory is stated for the network’s *structure*, determined consistently through a Laplace-approximation-based marginal posterior inclusion probability.

Quotient space. The strongest match in the Big Four. The paper defines an operator that maps any network to a canonical representative of its equivalence class under node permutation, sign changes and weight rescaling, and defines marginal posterior inclusion probabilities on that quotient. If you read one paper from this document, read this one.

Relevance tier: A — Direct

2023–2026 — no qualifying article identified

Journal of the Royal Statistical Society, Series B

2020–2021 — no qualifying article identified

2022

Sun, Y., and Liang, F. (2022). “A Kernel-Expanded Stochastic Neural Network.” *JRSS-B*, **84**(2), 547–578.
doi:10.1111/rssb.12496

Summary. Introduces the K-StoNet, which places support vector regression in the first hidden layer — mapping inputs into an infinite-dimensional feature space so that all local minima of the loss surface are globally optimal — and recasts the remaining network as a latent-variable model trained by imputation-regularized optimization.

Non-identifiability link. Never framed as “non-identifiability,” but the pathologies it targets — local traps and unreliable uncertainty quantification — are the downstream symptoms of a degenerate, symmetry-riddled parameter space; the reformulation sidesteps the degeneracy rather than quotienting it out.

Relevance tier: C — Adjacent (a workaround, not a study of the quotient)

2023–2026 — no qualifying article identified

Biometrika

2020

Song, Q., Sun, Y., Ye, M., and Liang, F. (2020). “Extended stochastic gradient Markov chain Monte Carlo for large-scale Bayesian variable selection.” *Biometrika*, **107**(4), 997–1004.
doi:10.1093/biomet/asaa029

Summary. Develops extended stochastic-gradient MCMC for sampling posteriors in large-scale Bayesian variable-selection problems, supplying the computational engine later used for StoNet-type and sparse-deep-learning models.

Non-identifiability link. Only indirect: it addresses the sampling difficulty created by multimodal, weakly-identified posteriors rather than studying non-identifiability itself. Include it for the computational lineage; drop it if you want strict topical matches only.

Relevance tier: C — Adjacent (borderline; computational lineage)

2021–2026 — no qualifying article identified

The Annals of Statistics

2020–2026 — no Tier A or Tier B article identified

This null result is worth stating plainly, because it is informative rather than a gap in the search. The prominent *Annals* deep-learning papers of this period are **frequentist** minimax-rate results, not Bayesian, and do not engage non-identifiability as a quotient: Schmidt-Hieber (2020, **48**(4), 1875–1897) on ReLU networks with sparse connectivity, and Kohler & Langer (2021, **49**(4), 2231–2249) on convergence rates for fully connected DNN regression estimates. Note also that the posterior-concentration result for sparse deep learning that readers often expect to find here — Polson & Ročková — was published at **NeurIPS 2018**, not in the *Annals*.

One genuine boundary case does exist and is listed for completeness:

Ohn, I., and Lin, L. (2024). “Adaptive variational Bayes: Optimality, computation and applications.” *Annals of Statistics*, **52**(1), 335–363.
doi:10.1214/23-AOS2349

Summary. Proposes an adaptive variational Bayes framework that computes a variational posterior on each candidate model and aggregates them by weights, attaining optimal contraction rates adaptively under general conditions, with deep neural network model spaces as a headline application.

Non-identifiability link. Genuinely Bayesian and genuinely about DNNs, but non-identifiability is not its subject: the model-selection machinery operates over an index of architectures, not over the symmetry quotient within a fixed architecture.

Relevance tier: Boundary case — Bayesian + DNN, but outside the non-identifiability theme

Summary matrix

Journal	Year	Article (first author)	Tier
<i>Biometrika</i>	2001	Paige & Butler — Bayesian inference in neural networks	B
<i>JASA</i>	2018	Liang, Li & Zhou — BNNs for drug sensitive genes	B
<i>JASA</i>	2022	Sun, Song & Liang — Consistent sparse deep learning	A
<i>JRSS-B</i>	2022	Sun & Liang — Kernel-expanded stochastic neural network	C
<i>Biometrika</i>	2020	Song, Sun, Ye & Liang — Extended SGMCMC	C
<i>Annals of Statistics</i>	—	No Tier A or B article identified, 2020–2026	—

The quotient-space thread specifically: the only Big Four article that builds explicitly on the equivalence-class-of-networks construction is Sun, Song & Liang (2022, *JASA*); the only one that ties posterior behaviour to non-identifiability by name is Paige & Butler (2001, *Biometrika*). Everything matching the modern description clusters around Faming Liang's group at Purdue.

Appendix — where the rest of this literature actually lives

If the goal is a working reading list rather than a Big Four audit, the density of relevant work is far higher just outside the four journals. Three directions are worth taking:

1. Machine-learning venues. Permutation symmetry, mode connectivity, and posterior sampling on the quotient are active NeurIPS / ICML / AISTATS / JMLR topics. Polson & Ročková (NeurIPS 2018) on posterior concentration for sparse deep learning is the natural companion to the *JASA* 2022 paper.

2. Second-tier statistics journals. *Bayesian Analysis*, *Annals of Applied Statistics*, *Electronic Journal of Statistics*, *Statistical Science*, and *JCGS* carry substantially more Bayesian-DNN identifiability work than the Big Four do.

3. Closing the completeness gap. To convert this from a substantiated survey into a guaranteed-exhaustive one, run a full-text search across each journal's archive for *identifiab**, *equivalence class*, *permutation invarian**, *weight-space symmetry*, *label switching*, and *quotient*, each intersected with *neural network*, then screen forward and backward citations of Sun, Song & Liang (2022), which is the hub node of this small literature.

A note on vocabulary that will improve recall: statisticians working on this problem more often write *label switching* — the term inherited from mixture models — than *quotient space*. Searching the former will surface papers the latter misses.

Compiled 23 July 2026. All bibliographic details verified against publisher records (Oxford Academic, Taylor & Francis, Project Euclid). Relevance tiers are the compiler's editorial judgement, not a formal classification.